John "Jack" P. McGuire

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EDUCATION

University of Wyoming, Laramie, WY
Ph.D. Student in Physics (Astronomy Concentration) August 2026–Present

Georgia State University, Atlanta, GA

- **Bachelor of Science in Physics** May 2026
  Overall GPA: 3.76; Institutional GPA: 3.81
- **Associate of Science in Physics, High Honors** May 2024
  Overall GPA: 3.73; Institutional GPA: 3.82

ACADEMIC EMPLOYMENT

University of Wyoming, Department of Physics & Astronomy August 2026–Present
Graduate Teaching Assistant

- Support undergraduate physics and astronomy instruction through laboratory or recitation teaching, grading, office hours, and individual student assistance.

PUBLICATIONS & MANUSCRIPTS

- Bentz, M. C.; Alahakone, R.; Azoulay, L.; Burns-Kaurin, E.; Chaturvedi, A. S.; Davis, M.; Kimani-Stewart, K.; Markham, M.; **McGuire, J. P.**; McMullan, E.; Patel, M.; Binu Raj, M.; Sankey, D.; Wilson, A. A.-L.; Zdanky, K. (2025). "Broad-band Monitoring of 797 Montana." *Minor Planet Bulletin*, **52**(3), 207–208.
- **Where Are the K & M Barium Dwarfs?** Manuscript in preparation; planned submission to an American Astronomical Society journal in Fall 2026.

SELECTED PRESENTATIONS & POSTERS

- **Finding Barium Dwarfs Among Established White-Dwarf Systems.** Poster presented at the Society of Physics Students Research Conference, April 23, 2026.
- **Finding Barium Dwarfs Among Established White-Dwarf Systems.** Poster presented at the Georgia State Undergraduate Research Conference (GSURC), April 6, 2026. Presented spectroscopic abundance-ratio measurements for five candidate barium dwarfs.
- **Posters at the Georgia State Capitol.** Selected to represent the Georgia State University Department of Physics & Astronomy and presented research on the CHARA Array and the gLOWCOST muon-monitoring network to legislators and the public, February 19, 2026.
- **AGB Post-Mass-Transfer Spectroscopic Ratios as White Dwarf Tracers.** Poster presented at the Georgia Regional Astronomy Meeting, Emory University, November 7–8, 2025.
- **Where Are the K & M Barium Dwarfs? Potential Answers in the 40 Eridani System.** Talk presented at the Georgia State University Undergraduate Research Symposium, August 7, 2025.

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Georgia State University August 2025–August 2026
Mentor: Dr. Russel J. White

- Investigated the 71 spectroscopically confirmed barium dwarfs using Gaia DR3 astrometry, radial-velocity statistics, archival spectroscopy, spectral-energy-distribution fitting, and MIST evolutionary models.
- Developed Python workflows using NumPy, Pandas, Astropy, Astroquery, and Matplotlib to cross-match catalogs, distinguish dwarfs from evolved interlopers, and compare barium dwarfs with white-dwarf–main-sequence binaries.
- Collaborated with Dr. Richard O. Gray of Appalachian State University on a high-resolution spectroscopic search for s-process enrichment in GJ 166 C. The null result, together with the system's roughly 200-year orbit, constrained the separation required for efficient AGB mass transfer.
- Continued CHARA interferometric reductions and analysis of nearby binary M dwarfs; contributed results to manuscripts on barium dwarfs and M-dwarf radius inflation.

Raghavan Fellow, Georgia State University Undergraduate Summer Research Program May–August 2025

Mentor: Dr. Russel J. White

- Initiated parallel projects on barium-dwarf binary evolution and CHARA interferometric measurements of the GJ 166 system.
- Reduced four nights of MIRC-X and MYSTIC data; fitted uniform-disk and limb-darkened models, propagated uncertainties, and compared independent reduction pipelines.
- Identified the strong concentration of known barium dwarfs among F- and G-type stars, motivating a search for the missing K- and M-dwarf population.

Cosmic-Ray Research Assistant, Georgia State University 2025–2026
Mentors: Dr. Xiaochun He and Dr. Viacheslav Sadykov

- Assembled Raspberry Pi-based detectors for the gLOWCOST global muon-monitoring network and supported detector-based science outreach.
- Analyzed multi-year muon-flux measurements using atmospheric data, solar and lunar ephemerides, Fourier transforms, coherence analysis, and correlation tests to investigate environmental and tidal modulation.
- Studied Forbush decreases by comparing gLOWCOST measurements with ACE MAG and SWEPAM solar-wind and magnetic-field data from the L1 point.

OBSERVING EXPERIENCE

Center for High Angular Resolution Astronomy (CHARA) Array November 2025

- Planned and led two half-night observing sessions with PI Becky Flores using the six-telescope CHARA Array.
- Operated MIRC-X (H band) and MYSTIC (K band) to observe nearby M dwarfs in binary systems and measure their angular diameters.
- Reduced and modeled archival observations of GJ 166 C, finding an approximately 30% radius excess relative to stellar-evolution model predictions.

Hard Labor Creek Observatory February 2025

- Participated in a four-night photometric campaign on main-belt asteroid 797 Montana using the 24-inch Miller telescope.
- Reduced CCD photometry using IRAF and DS9; the resulting light curve yielded a rotation period of 4.545 ± 0.001 hr and led to a refereed publication.

PROFESSIONAL TRAINING & PUBLIC SERVICE

American Physical Society – Advocacy Champion June 2026–Present

- Participate in APS science-policy and advocacy training and coordinated outreach supporting research funding, STEM education, and the physics workforce.
- Build sustained relationships with elected officials and their staff to communicate the priorities and public value of the physics community.

Hello SciCom – Comedy Camp for Science August–October 2026

- Selected for an eight-week virtual training program in using humor, storytelling, and performance to communicate technical STEM research accurately and accessibly.
- Develop a three-minute public-facing talk and contribute to a research study examining how humor affects trust, engagement, and perceptions of science as a public good.

SKILLS & INTERESTS

Scientific Computing: Python (NumPy, Pandas, Matplotlib, Astropy, Astroquery, Specutils, Lightkurve), TOPCAT, LaTeX, HTML
Astronomy Tools: Gaia Archive/ADQL, MIST isochrones and mass tracks, VOSA spectral-energy-distribution fitting, IRAF, DS9, CHARA MIRC-X/MYSTIC reduction and interferometric model fitting
Instrumentation & Computing: Arduino, Geant4, LabVIEW, macOS, Windows, Linux
Research Interests: Binary evolution, chemically peculiar stars, stellar interferometry, stellar atmospheres, white dwarfs, astronomical data science, astrobiology, and science communication
Professional Affiliations: Sigma Pi Sigma; Society of Physics Students

PROFESSIONAL EXPERIENCE

Vertigo Pinball, LLC – Startup Consultant and Manager December 2020–July 2026

- Researched the barcade industry, prepared profit-and-loss projections, selected an arcade-machine lineup, and designed the proposed physical layout.
- Founded and managed the Georgia Pinball Museum initiative, curating interactive exhibits that served hundreds of guests daily during summers 2023 and 2024.
- Led Blue Ridge Gourmet Popcorn production, packaging, and specialty-flavor development from Summer 2024 through July 2026.

British Swim School – Swim Instructor & Substitute Spring 2022–Winter 2025

- Taught private and group lessons for students ranging from infants to older adults, emphasizing water safety, technique, confidence, and incremental goal-setting.

Huntcliff Club – Assistant Head Swim Coach Summer 2016–Summer 2021

- Coached swimmers ages 4–18, developed competitive lineups using TeamUnify, and helped build a strong team culture.

Camp-Run-A-Mutt Spring 2020–Spring 2022

- Cared for 5–25 dogs at a cage-free day and overnight boarding facility and coordinated closely with staff and pet owners.

LEADERSHIP & ACTIVITIES

Eagle Scout, Boy Scouts of America Troop 463 Awarded August 2020

- Planned and led a landscape redesign for Mount Vernon Presbyterian Church, including installation of hundreds of grass bundles and three Japanese maple trees.

Georgia State University Perimeter SPACE Club – Secretary Fall 2022–Spring 2024

- Coordinated elementary-school outreach using physics demonstrations and virtual-reality experiences of ISS experiments and operations.